

$$\begin{array}{ccccc}
 F^T U^T [\boldsymbol{M}, \boldsymbol{N}]_T & \xrightarrow{=} & F^T [M, N] & \xrightarrow{\psi_{M,N}} & [F^T M, F^T N]_T \\
 \downarrow \epsilon_{[M,N]_T} & & & & \downarrow [F^T M, \epsilon_N]_T \\
 & & (iv) & & [F^T M, \boldsymbol{N}]_T \\
 & & & & \uparrow [\epsilon_M, N]_T \\
 [\boldsymbol{M}, \boldsymbol{N}]_T & \xrightarrow{=} & & & [\boldsymbol{M}, \boldsymbol{N}]_T
 \end{array}$$